

The Ancients' Sky: The Three Enclosures and the Twenty-Eight Mansions

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Abstract

In 1978, at Leigudun in Suixian, Hubei (now Suizhou), an archaeological team opened a Warring States tomb. Its occupant was Yi, a ruler of the state of Zeng whom later ages call Marquis Yi of Zeng. Beside its famous set of chime bells, the tomb yielded a lacquered wooden clothes chest, catalogued E.66. On its lid, around a large seal-script character for the Dipper, are written twenty-eight star names, with a dragon drawn to the east and a tiger to the west. This note uses that object as its starting point and sets out the two schemes by which the ancients divided the sky. The first is the twenty-eight mansions (二十八宿), star regions laid out one to each night of the moon's eastward circuit, along the ecliptic and near the celestial equator; the note explains what a mansion is, lists the twenty-eight names in their fixed order in four groups of seven, and records the four guardian animals — Azure Dragon, Dark Warrior, White Tiger and Vermilion Bird — showing that their left-and-right arrangement follows from the single posture of observing while facing south. The second scheme is the three enclosures (三垣): the Purple Forbidden, Supreme Palace and Heavenly Market Enclosures around the north celestial pole. The note then explains how the grid was used, at dusk and at dawn, to record position and to infer the season. The note is descriptive; it contains no predictive claims.

Keywords: twenty-eight mansions; three enclosures; four images; celestial equator; ecliptic; determinative star; Zeng Hou Yi lacquer chest; Chinese astronomy

1. The character su comes first

In 1978, at Leigudun in Suixian, Hubei — now Suizhou — an archaeological team opened a tomb of the Warring States period.

The occupant's name was Yi, a ruler of the state of Zeng whom later ages call Marquis Yi of Zeng. The tomb is best known for its set of chime bells. From the same tomb was also recovered a lacquered wooden clothes chest, catalogued E.66.

According to the inscription in the chest, it was originally called hu (匱) and was used for storing clothes. The lid, taken off, is covered with a layer of black lacquer, on which a diagram is drawn in vermillion.

At the centre of the diagram is a large character 斗 (dou, the Dipper) in seal script. Written around it in a circuit are twenty-eight star names. To the east a dragon is drawn, to the west a tiger.

This is the earliest physical object bearing the twenty-eight mansions that can be seen today. The burial of Marquis Yi of Zeng is generally dated to around 433 BCE. That is to say, as early as the fifth century BCE these twenty-eight names were already common knowledge, casually painted on the lid of an everyday chest.

Where those twenty-eight names came from, and what use the division served — this must begin with the character 宿 (su).

Su (宿) is the su of lodging for the night.

The moon moves through the starry sky, shifting eastward by about thirteen degrees a day; each day it passes into a different region of the sky. One region per night, like a traveller changing inns each day. Following this rhythm, the ancients divided the stars near the ecliptic and the celestial equator in turn into twenty-eight star regions, each region called a mansion (宿, su).

The twenty-eight mansions are also called the twenty-eight lodgings or the twenty-eight stars.

Wang Chong of the Eastern Han has a sentence in the *Lunheng* that puts the matter plainly:

二十八宿为日月舍，犹地有邮亭，为长吏廨矣。(The twenty-eight mansions are the lodgings of the sun and moon, just as on the earth there are postal stations, which are the public offices of the senior officials.)

The mansions in the sky are the relay stations of the sun and moon, comparable to the postal stations on earth, the official buildings where passing officers rest.

This sentence names the use of the twenty-eight mansions: recording position. Where the sun, moon and the five planets have reached, the ancients recorded with a phrase such as "entering such-and-such a mansion by so many degrees". Within each mansion a star was chosen as a marker, called the **determinative star** (距星, *jùxīng*), and positions were measured from it. Spread the twenty-eight cells across the sky, and the sky acquires a scale.

The earlier notes dealt with scales on the ground: how the year comes about, how the month, how the day is cut, how the double-hour is divided. The twenty-eight mansions are the set in the sky. Only the two sets together make a complete base map.

In the "Spring Offices" section of the *Zhouli* there is an office called the *Fengxiang shi*, whose charge was precisely the positions of the year, the month, the nodes, the day and the twenty-eight stars. The number appears in the classics very early.

2. The twenty-eight names

The twenty-eight mansions have a fixed order, beginning with the Horn and running from west to east, in the direction of the moon's motion. Each of the four directions has seven mansions, and each direction has a name:

Direction	Name	The seven mansions
East	Azure Dragon (Canglong)	Horn · Neck · Root · Room · Heart · Tail · Winnowing Basket
North	Dark Warrior	Dipper · Ox · Girl · Emptiness · Rooftop · Encampment · Wall
West	White Tiger	Legs · Bond · Stomach · Hairy Head · Net · Turtle Beak · Three Stars
South	Vermilion Bird	Well · Ghost · Willow · Star · Extended Net · Wings · Chariot

Several of these characters are awkward to read, so their sounds are noted first: 亢 is read kàng, 氐 dī, 箕 jī, 娄 lóu, 昂 mǎo, 觜 zī, 参 shēn, 轸 zhěn.

There are two "Dipper" characters, easily confused. The seven stars of the Northern Dipper are in the northern sky, the ladle that scoops wine; the Dipper mansion is in the southern sky with only six stars, called the **Six Stars of the Southern Dipper**. The two "Dippers" face each other across the celestial pole, and poets often set the two against each other.

Among the twenty-eight mansions, four can still be recognised today by looking up:

The core of the Heart mansion is called Antares (心宿二); the old books call it **Great Fire** (大火, *dàhuo*); it is reddish in colour and one of the brightest stars in the whole sky.

The Hairy Head mansion is a small knot of stars; a person with good eyesight can make out six or seven crowded together. This is the Pleiades cluster, which the West calls the Seven Sisters.

The Three Stars mansion corresponds to today's Orion; its three central stars are strung in a horizontal line and are easiest to recognise on a winter night.

The Dipper mansion is the Six Stars of the Southern Dipper just mentioned.

3. Four animal names, from one posture

The seven eastern mansions are the Azure Dragon, the seven western the White Tiger, the seven southern the Vermilion Bird, the seven northern the Dark Warrior. The four animals divide the twenty-eight mansions into four parts, together called the four images (四象). The Liji, "Qu li, part one", has a line:

行前朱鸟而后玄武，左青龙而右白虎。(In marching, the Vermilion Bird in front and the Dark Warrior behind, the Azure Dragon on the left and the White Tiger on the right.)

Kong Yingda of the Tang, subcommenting on it, says plainly:

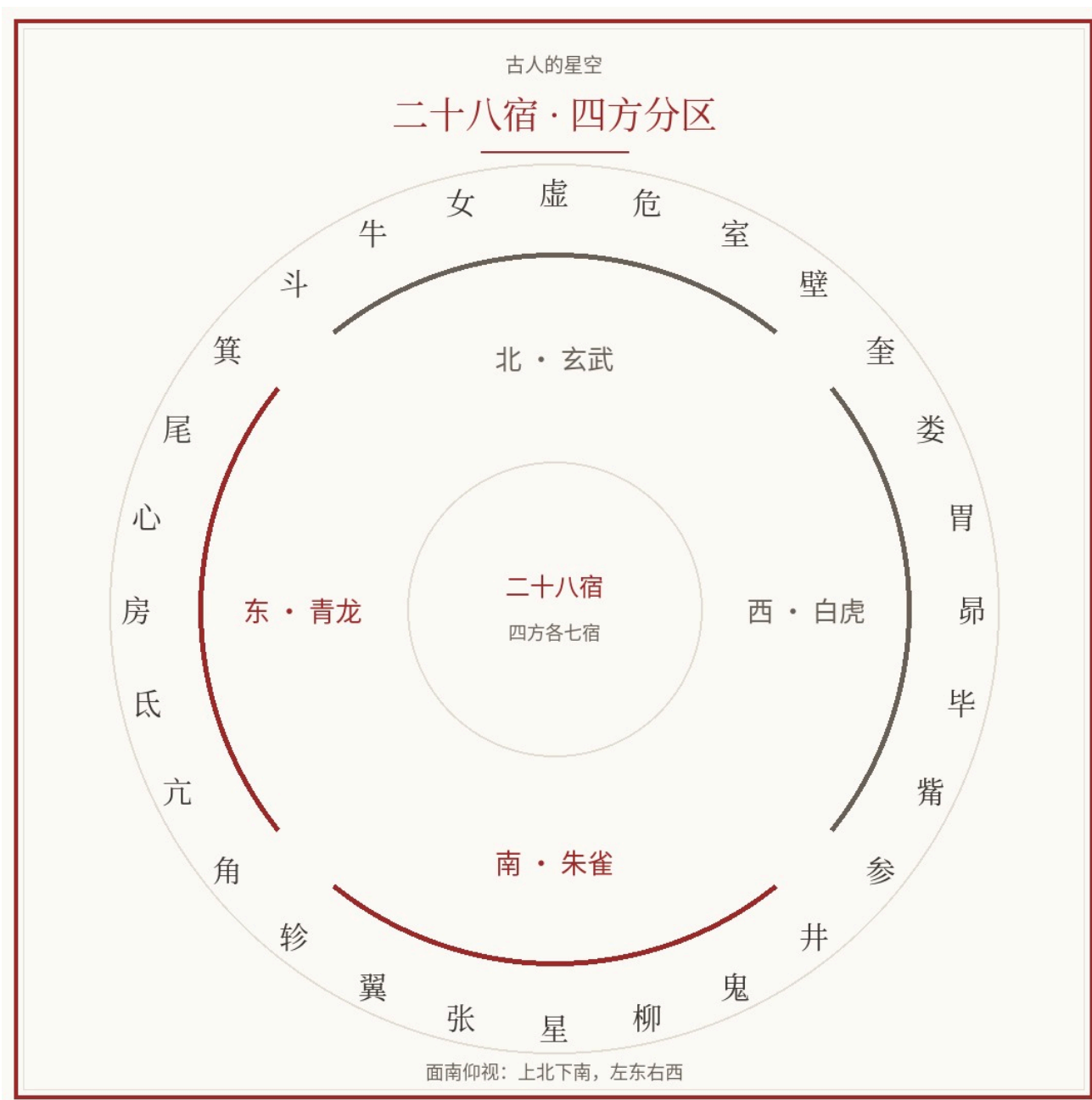
朱鸟、玄武、青龙、白虎，四方宿名也。(The Vermilion Bird, the Dark Warrior, the Azure Dragon and the White Tiger are the names of the mansions of the four directions.)

The names of the mansions of the four directions. What these four names denote is the four directional divisions of the twenty-eight mansions, not the four animals themselves. By the way, the Liji writes the Vermilion Bird as 朱鸟 (zhuniao); later ages mostly write it 朱雀 (zhuque); the same group is meant.

Then where do the left and right in "Azure Dragon on the left, White Tiger on the right" come from?

From a single action: standing facing south.

The ancients were accustomed to watching the stars facing south. At dusk around the spring equinox, standing facing south, overhead is the southern sky, where the Vermilion Bird is; on the left hand is the eastern sky, where the Azure Dragon has just risen; on the right hand is the western sky, where the White Tiger is half hidden below the horizon; behind is the northern sky, where the Dark Warrior has already set. The four directions and the four animals are all derived from the single posture of "observing while facing south". Change the direction you stand facing, and left and right are thrown into confusion.



Some scholars doubt that so neat a pairing of the four images with the twenty-eight mansions was a later addition, made after the Han. But since the dragon, the tiger and the twenty-eight mansions share a single lid on the lacquer chest, it does not hold that the pairing had no basis at all.

4. There are also three "cities" in the sky

The twenty-eight mansions govern a belt of sky along the ecliptic and the celestial equator. The region near the north celestial pole has another scheme of division, called the **three enclosures** (三垣, *san yuan*).

Yuan (垣) means a wall. The three enclosures are three walled cities.

The Purple Forbidden Enclosure rings the north celestial pole and was taken by the ancients as the dwelling of the celestial emperor; the pole star and the Dipper are both within it. The Supreme Palace Enclosure lies south of the Purple Forbidden, the imperial court, its star names mostly official titles. The Heavenly Market Enclosure lies further south, a marketplace, its star names including a crowd of "rows of shops", "rows of carts", the Dipper and the grain measure, all implements of trade and of measuring.

The three enclosures together with the twenty-eight mansions are, in the eyes of the ancients, the whole of the starry sky. The scheme took its roughly final form in the Sui and Tang, and can be traced further back. The Forbidden City in Beijing was laid out with reference to the Purple Forbidden Enclosure, holding to the principle of "setting out the capital in the image of heaven".

The preceding note, on time, mentioned that the earth's axis of rotation slowly traces a circle in space. The position of the pole star shifts, and the "pole star" is not forever fixed on one particular star. That matter is left to the note on precession. Here one sentence need only be kept in mind: the *Kaogong ji*, speaking of fixing direction at night, uses the five characters 夜考之极星, "at night examine the pole star". That pole star is in the Purple Forbidden Enclosure.

5. How the grid is used

The twenty-eight mansions are not there for show. The ancients used them chiefly to watch four moments:

A mansion just rising in the east at dusk is called "dusk rising"; a mansion hanging due south at dusk is called "dusk culmination"; rising in the east before dawn is "dawn rising"; hanging due south before dawn is "dawn culmination".

The four moments are four rulers. A given mansion reaches due south at dusk only during a certain stretch of the year. Record them day by day, and the season at hand can be inferred backwards.

Set this aspect aside for now; the note devoted to the solar terms will take up "which mansion is due south, and what farm work should be done".

One further aspect of the twenty-eight mansions is to be used later: direction.

How is one to tell east, west, north and south at night? All the stars in the sky are turning; only one place barely moves. Find that place, and direction is had.

Closing

Return to that lacquer chest.

The twenty-eight characters on the chest lid can still be made out one by one by a person today. In the two thousand four hundred years since, the positions of the stars have shifted a good deal, yet the names and the order of this set of cells have scarcely changed.

The ancients drew a grid on the sky in order to know where they stood. Look up, and a certain mansion is directly overhead; direction is had.

What follows in this series is that matter. In the years and months before the compass, how a pole, two shadows and one unmoving star measured out east, west, north and south.

A&Q

What are the twenty-eight mansions?

The twenty-eight star regions divided out along the ecliptic and near the celestial equator, each region called a mansion. Wang Chong of the Eastern Han, in the *Lunheng*, says "the twenty-eight mansions are the lodgings of the sun and moon" — the relay stations of sun and moon.

How are the twenty-eight mansions arranged?

From the Horn, running from west to east, in the direction of the moon's motion. Seven in each of the four directions, matched with the Azure Dragon, the Dark Warrior, the White Tiger and the Vermilion Bird.

In "Azure Dragon on the left, White Tiger on the right", from which side are left and right reckoned?

Standing facing south. Facing south, the left hand is the eastern sky, the right hand the western sky, and behind is the northern sky.

What are the three enclosures?

The three "cities" near the north celestial pole — the Purple Forbidden Enclosure, the Supreme Palace Enclosure and the Heavenly Market Enclosure — arranged from north to south. Together with the twenty-eight mansions they make the whole of the starry sky as the ancients saw it.

What use are the twenty-eight mansions?

Recording position, and also reading the seasons. Watch a mansion's dusk rising, dusk culmination, dawn rising and dawn culmination, and the season at hand can be inferred backwards.

Note by Chujia Kuang

The string of names of the twenty-eight mansions looks at first like a heap of unrecognisable characters. Read smoothly, one sees that it is in fact a table — the sky divided into twenty-eight

cells, so that the positions of sun, moon and the five planets have something to be stated against. Looking up at the sky, what the ancients first understood was this: everything in the sky is moving, but it moves by rule, and can be recorded and inferred. The calendar grew out of that rule. This is groundwork; later, in treating the solar terms and precession, we shall return to this table.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the 史记: 愚者千虑, 必有一得, "the foolish man's thousand thoughts must yield one gain". Where something is written wrongly, corrections are welcome.

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